Jiayun JIN

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PROFILE

Quantitative researcher and survey methodologist with a PhD in survey methodology and strong experience in measurement, survey data quality, statistical modeling, and communicating quantitative findings across technical and non-technical audiences. I am interested in building reliable measurement systems that turn survey responses, behavioral signals, and analytical evidence into clear insights for product and business decisions.

CORE STRENGTHS

- **Survey and experience measurement:** expertise in questionnaire-based measurement, response quality, measurement error, latent constructs, experience and attitude measurement, and interpretation of survey indicators.
- **Quantitative modeling:** experience with large-scale survey data, statistical modeling, latent-variable models, mixture models, weighting, and simulation-based method evaluation.
- **Insight communication:** extensive experience translating complex statistical findings into clear explanations through teaching, advising, reports, presentations, and invited talks.
- **Reproducible analytics:** advanced R skills for survey data analysis and statistical modeling; active use of Python, Git/GitHub, Markdown, and documented analytical pipelines.
- **Collaborative research delivery:** experience contributing to survey design, leading funded research projects, supervising applied analytical work, and working across statistics, survey methodology, sociology, and computational social science.

EMPLOYMENT

2022–2025 **Assistant Professor**, Department of Sociology and Demography, Renmin University of China

- Contributed to the Chinese General Social Survey (CGSS), a large-scale national probability survey in China, participating in questionnaire design and survey program discussions.
- Designed and led survey-based research projects on attitudes, response behavior, online surveys, and data quality, covering the full research cycle from measurement design to analysis, interpretation, and reporting.
- Led a National Social Science Fund of China project on measurement error and data quality in online surveys in algorithmic data environments, focusing on reliability of survey responses and digital data collection processes.
- Developed and applied statistical models to evaluate what survey responses measure, how response processes affect data quality, and how observed indicators relate to underlying

constructs, including published work using CGSS gender-attitude data.

- Taught quantitative research methods, statistics, and data analysis with R; supervised student projects on AI-assisted survey design, online response behavior, and computational analysis of social data.

2020–2021 **Research Assistant**, Center for Survey Methodology, KU Leuven

- Developed statistical process monitoring methods for assessing survey data collection quality and identifying irregular response patterns during large-scale survey fieldwork.
- Built reproducible monitoring tools and reporting workflows for evaluating response quality indicators and communicating data quality signals during ongoing survey data collection.
- Worked with survey methodology experts on data quality assessment, measurement error detection, and communication of monitoring results to research and fieldwork stakeholders.
- Presented this work at the NPSO Innovatiedag, hosted at Statistics Netherlands (CBS), and received the **Innovation Prize**.

SELECTED ANALYTICAL PROJECTS

Non-Probability Survey Inference github.com/jinjiayun57/nonprob-survey-inference
R-based simulation workflow for evaluating survey bias and measurement assumptions. 2026, active

- Built a reproducible R workflow to compare calibration weighting, inverse probability weighting, mass imputation, and doubly robust estimation under different selection mechanisms and auxiliary-information conditions.
- Documents assumptions, estimator behavior, and interpretation in a transparent GitHub workflow.

OpenCodebook github.com/jinjiayun57/open-codebook
Prototype for transparent LLM-assisted coding of open-ended survey responses. 2026, active

- Developed a Python-based workflow in which researchers define structured codebooks and local open-source language models apply codes to open-ended text responses.
- Includes uncertainty flags and audit trails to make interpretive coding decisions more transparent and reviewable.

TECHNICAL SKILLS

Quantitative research and survey analysis

- Survey methodology; questionnaire-based measurement; response quality; measurement error; survey indicator interpretation; experience and attitude measurement.
- Statistical modeling for survey and attitudinal data; latent-variable modeling; mixture modeling; response-style analysis.

- Bias assessment, weighting, non-probability sample inference, and simulation-based method evaluation.

Programming, data, and communication

- **R**: advanced statistical modeling, survey data analysis, simulation studies, data cleaning, visualization, and reproducible reporting with R Markdown / Quarto.
- **Python**: pandas, NumPy, scikit-learn, Jupyter notebooks; applied machine learning and text-analysis workflows.
- Git/GitHub, documented analytical pipelines, Markdown, LaTeX, Jupyter; familiarity with large-scale structured datasets and SQL-based data environments.
- Strong experience translating complex quantitative results into clear written, visual, and oral explanations for non-technical audiences.

EDUCATION

2015–2021	**Ph.D. in Social Sciences**, KU Leuven, Belgium — *Centre for Sociological Research* Dissertation: *The Use of Control Charts to Monitor and Manage Survey Data Collection Process from the Perspective of Measurement Errors*
2011–2013	**M.Sc. in Statistics**, Stockholm University, Sweden
2009–2011	**M.A. in Statistics**, Renmin University of China
2005–2009	**B.A. in Statistics**, Xiamen University, China

SELECTED PUBLICATIONS

Jin, Jiayun. (2025). Disentangling and Reassembling: Unveiling Hidden Patterns in Gender Attitudinal Surveys. *Sociological Methods & Research*. `doi:10.1177/00491241251413678`

Jin, Jiayun, & Loosveldt, Geert. (2022). Nonparametric Multivariate Control Charts for Numerical and Categorical Variables. *Communications in Statistics – Simulation and Computation*, *53*(1), 457–475. `doi:10.1080/03610918.2021.2023572`

Jin, Jiayun, & Loosveldt, Geert. (2021). Identifying Outliers in Response Quality Assessment by Using Multivariate Control Charts Based on Kernel Density Estimation. *Journal of Official Statistics*, *37*(1), 97–119. `doi:10.2478/jos-2021-0005`

Jin, Jiayun, & Loosveldt, Geert. (2020). Assessing Response Quality by Using Multivariate Control Charts for Numerical and Categorical Response Quality Indicators. *Journal of Survey Statistics and Methodology*, *9*(4), 674–700. `doi:10.1093/jssam/smaa012`

Jin, Jiayun, Vandenplas, Caroline, & Loosveldt, Geert. (2019). The Evaluation of Statistical Process Control Methods to Monitor Interview Duration during Survey Data Collection. *SAGE Open*, *9*(2). `doi:10.1177/2158244019854652`

SELECTED PRESENTATIONS AND RECOGNITION

2025 **Invited Talk, China Media Group, Beijing**
Presented survey error sources and their implications for media professionals; translated statistical concepts for a non-technical professional audience.

2024 **Web Survey Frontier Forum, Renmin University of China, Beijing**
Presented measurement error in web surveys and non-sampling error in online response behavior.

2019 **Innovation Prize, NPSO Innovatiedag**
Netherlands Platform for Survey Research, hosted at Statistics Netherlands (CBS), for applied work on survey data quality monitoring.

2024 **Young Faculty Teaching Skills Award**
Renmin University of China

2017, 2019 **European Survey Research Association Conference**
Presented research on statistical process control and response behavior in survey data collection.

LANGUAGES

English (fluent) · Chinese (native) · Dutch (basic, actively learning)